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Zixu Wang

MSc Artificial Intelligence • Lancaster University

Profile

MSc Artificial Intelligence student at Lancaster University (79.4% distinction average), working on explainable, multimodal and agentic AI. My work turns model behaviour into auditable, statistically validated evidence and reproducible pipelines across explainable medical AI, multi-agent LLM systems and NLP, with experience leading research teams of up to five members.

Education

Lancaster University — Lancaster, UK
MSc Artificial Intelligence — **Sep 2025 – Sep 2026**
Average-to-date: 79.4% (Distinction, indicative). Modules: Machine Learning, NLP and Language Models, Intelligent Agents and Autonomous Systems, Advanced Topics in AI, Data Science and AI Fundamentals.

Aston University — Birmingham, UK
BSc Computer Science (Hons) — **Sep 2021 – Jul 2024**
Second Class Honours. Selected modules: Mathematics (79.0), Human–Computer Interaction (70.0), Team Project, Internet Applications & Database Design, Image & Video Processing.

Research

MSc Dissertation • Toward Explainable Face2Gene Diagnosis — *Jun 2026 – Sep 2026*
Lancaster University, supervised by Dr. Richard Jiang | NumPy • XAI • Medical AI

- **Context:** Explainable medical-AI dissertation — facial-phenotype diagnosis models behave as black boxes, blocking clinical trust on low-resolution 32×32 face images.
- **Role:** Sole researcher; independently designed the method, pipeline and full experimental protocol.
- **Result:** A reproducible coarse-ROI Anatomical Evidence Vector (AEV) pipeline (NumPy MLP on Face2Brain / PD-DBS) converting occlusion behaviour into quantitative ROI-level evidence, audited via QC, calibration, FDR-corrected statistics, leakage/confounding and robustness checks.

Neglected Tropical Disease Clinical Trials Analysis — *Sep 2025 – Dec 2025*
SCC.450 Introduction to AI and Data Science | Python • Regression • Network Analysis

- **Context:** Course research project (final report 77.5%) quantifying global inequity in neglected tropical disease clinical research.
- **Role:** Team lead (2–3 members) — coordinated the data pipeline and analysis split, and owned the statistical modelling.
- **Result:** Cleaned 315 WHO ICTRP trial records across 62 countries; built logistic-regression models, a sponsor–country–disease network analysis and publication-quality visualisations.

Projects

Diversity Beats Strength? Multi-LLM Teams in Game-Theoretic Settings — *LLM Agents • Multi-Agent • Simulation*
SCC.452 Intelligent Agents — group project: 86%

- **Led a 4–5 person team**, designing a modular multi-LLM evaluation architecture to test whether diverse mixed-agent teams outperform homogeneous strong-model teams.
- Ran approx. 6,400 trial simulations across Prisoner's Dilemma, Snowdrift and Stag Hunt with GPT-4o, DeepSeek-V3 and Gemini-3-Flash, analysing cooperation, payoff, cheap-talk, memory, network topology and team composition.

Image Classification & Clustering — *PyTorch • DINOv2 • Clustering*
SCC.451 Machine Learning

- Applied transfer learning and fine-tuning with ResNet50, DenseNet121 and DINOv2 for image classification; used K-means, hierarchical clustering and PCA for climate-data analysis.

Sentiment Analysis across Multilingual Datasets — *NLP • English/Chinese*
SCC.453 NLP and Language Models

- Compared tokenisation, stemming and lemmatisation strategies for English and Chinese sentiment-classification pipelines, evaluating preprocessing impact on accuracy.

Undergraduate Projects: Restaurant Back-Office System & PC Components E-Commerce Site — *Aston University*
Individual / Team

- Built a restaurant backend/admin management system (final-year individual project) and a team-based e-commerce site for PC components, contributing to requirements, implementation, testing and delivery.

Technical Skills

- **Programming:** Python, SQL, Java
- **AI / ML:** PyTorch, scikit-learn, NumPy MLPs, transfer learning, clustering, classification, model evaluation
- **NLP / LLMs:** Text preprocessing, OpenAI / Anthropic / Google API integration, agent simulations
- **Data / XAI:** pandas, NumPy, matplotlib, seaborn, NetworkX, occlusion analysis, calibration, AUROC/AUPRC, FDR-corrected statistics
- **Tools & Languages:** Git, Linux, Jupyter, LaTeX; Chinese (native), English (fluent)